Name: Madhulatha Reddy Gurrala

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Domain: AI for Finance

KisanFin: Project SOP

Project Title: KisanFin: Smart Credit Scoring for Farmers and Gig Workers

Domain: AI for Finance (Domain 4)

Problem Statement: Millions of small farmers and gig workers have no credit history and irregular incomes that traditional credit models cannot assess. Banks reject them or they borrow from moneylenders at very high interest. Low literacy and English-only apps make it harder for them to access formal credit.

Target Users: Small farmers, gig workers (delivery and ride platforms), and loan officers at banks and NBFCs.

Proposed Solution: A web app that predicts default risk from alternative data (UPI activity, mobile recharge regularity, bill payments, income pattern, land or platform tenure) and returns a 0 to 100 score, a suggested loan limit, plain-language reasons, and tips to improve, in English and Telugu.

Key Features:

- Credit score (0 to 100) with risk gauge
- Suggested loan limit
- Explainable reasons for each score
- English and Telugu language toggle
- “How to improve your score” tips
- Separate inputs for farmers and gig workers

Technology Stack:

- Frontend and backend: Streamlit (Python)
- AI model: XGBoost
- Data: synthetic CSV dataset (5,000 records)
- Libraries: Pandas, NumPy, scikit-learn, Plotly
- Source control: GitHub
- Deployment: Streamlit Community Cloud

AI Usage:

- XGBoost predicts the probability of default (AUC 0.741 on test data).
- Feature contributions from XGBoost’s built-in TreeSHAP method explain each score.

- An AI coding assistant helped with brainstorming, code, debugging and documentation. All code was reviewed and tested by the team.

System Workflow:

1. User selects Farmer or Gig worker and enters details.
2. The model predicts the default probability.
3. The probability is converted to a 0 to 100 score and a loan limit.
4. Feature contributions are turned into plain-language reasons and tips in the chosen language.
5. The result is shown with a gauge, loan limit, risk and reasons.

Security and Privacy:

- No API keys, passwords or secrets in the code or repository.
- Demo uses synthetic data only, and user inputs are not stored.
- Real deployment would use consented data through Account Aggregator, with fairness audits across gender and age.
- The score is decision support, not a final lending decision.

Testing Plan:

- Strong profile should score high and weak profile should score low.
- Test the language toggle, and farmer versus gig worker inputs.
- Test boundary values for income, dependents and age.
- Test the live link on a phone, another browser and Incognito (done).

Deployment Plan: Push code to a public GitHub repository, deploy from GitHub on Streamlit Community Cloud, and test the live URL on another device.

GitHub: https://github.com/madhureddy24/KisanFin-AI-Credit-Scoring

Live app: https://kisanfin-ai-credit-scoring-iybmdxcbufcmwv4cyvymjt.streamlit.app/

Research Summary:

- **Existing approaches:** Traditional bureau scores rely on past loan history, so first-time borrowers get no score. Government schemes help farmers but need paperwork and bank visits. Some fintech lenders use alternative data, but few explain decisions or support regional languages.
- **Gap:** There is little explainable, regional-language credit scoring for irregular-income workers.
- **Our contribution:** An explainable, bilingual scoring prototype with actionable improvement tips.